

Local Continuum Yang–Mills Limit from Particle Trajectories

Abstract

This paper is a self-contained translation of the local continuum-limit proof stack from the earlier Fractal Set Yang–Mills chapter into the notation of the *4_ymmg* series. It is intended to serve as a *03b* replacement for the current lattice paper, but in theorem dependence it relies only on the convergence backbone of *01_convergence*. The finite QFT lattice, the $SU(3)$ link variables, the plaquette holonomies, the discrete field strengths, and the Wilson action are restated here from first principles. The continuum section then follows the old route in the same logical order: the flat refinement regime $a_N = \max\{\tau_N, \rho_N\} \downarrow 0$, the small-plaquette Wilson expansion, the area and counting control, the weighted Riemann-sum limit, the global remainder estimate, and the coupling identification that yields the Euclidean Yang–Mills action on $\mathbb{R}^4$.

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1 Purpose

The present paper is designed to substitute the current lattice paper at the point where the gauge-sector target is identified. Its role is narrower than that of a full new volume: it takes the old local continuum route, rewrites it in the *4_ymmg* notation, and makes the finite lattice definitions self-contained so that the only external theorem dependence is on *01_convergence*.

The paper therefore has two jobs. First, it records internally the full finite trajectory-induced QFT lattice and its associated $SU(3)$ gauge data:

$$\mathcal{L}^{(N)} = (\mathcal{N}^{(N)}, \mathcal{E}^{(N)}), \quad \Xi^{(N)} : \mathcal{N}^{(N)} \rightarrow \mathbb{R}^4, \quad \mathcal{C}_{\square}^{(N)},$$

together with the finite link variables $U_3^{(N)}(e)$, plaquette holonomies $U_3^{(N)}(P)$, discrete field strengths $F_P^{(N)} = (ig_3 A_P^{(N)})^{-1} \log U_3^{(N)}(P)$, and Wilson action $S_W^{(N)}$. Second, it translates the old five-step continuum proof without changing its structure:

1. identify the lattice spacing by $a_N = \max\{\tau_N, \rho_N\}$;
2. expand the plaquette Wilson density for small plaquettes;
3. control plaquette areas and plaquette counts;
4. pass the weighted plaquette sum to a Riemann integral;
5. identify the $SU(3)$ normalization of the continuum action.

No bilocal thermodynamic target is used anywhere in the theorem chain of the present paper.

2 Where the old route takes the lattice spacing to zero

The old document uses two primitive geometric scales:

- a temporal step τ on worldline links;
- a spatial or localization scale ρ on interaction links.

In the gauge-sector continuum analysis those two scales are bundled into a common small parameter

$$a := \max\{\rho, \tau\},$$

and the continuum limit is the simultaneous regime

$$\rho \rightarrow 0, \quad \tau \rightarrow 0, \quad a \rightarrow 0.$$

The purpose of this section is to rewrite that old regime in the notation of the *4_ymmg* papers, where the particle number N already indexes the family of finite systems.

Definition 2.1 (Flat refinement family). A *flat refinement family* is a sequence of finite trajectory data sets

$$\mathcal{Q}^{(N)} = (\mathcal{L}^{(N)}, \Xi^{(N)}, \{U_3^{(N)}(e)\}_{e \in \mathcal{E}^{(N)}}, \mathcal{C}_{\square}^{(N)})_{N \geq N_0}$$

together with positive scales τ_N, ρ_N such that

$$a_N := \max\{\tau_N, \rho_N\} \downarrow 0 \quad \text{as } N \rightarrow \infty.$$

The quantity a_N is the continuum-limit lattice spacing in the new notation.

Remark 2.2. This is exactly the old $\rho, \tau \rightarrow 0$ route rewritten with the particle index N as the external family parameter. Nothing essentially new is introduced here: the only change is that the small parameter is now denoted by a_N rather than by a .

3 Trajectory-induced QFT lattice

We now restate the finite QFT lattice directly in this paper. This removes the need to cite any external document for the geometric lattice construction on which the continuum route is built.

Definition 3.1 (Finite parameters). Fix once and for all the positive parameters

$$\kappa_0 > 0, \quad \nu_{\text{visc}} \geq 0, \quad g_3 > 0, \quad \ell_0 > 0, \quad \hbar_{\text{eff}} > 0, \quad \mu_p > 0, \quad \epsilon_c > 0.$$

For each N , the temporal scale $\tau_N > 0$ and interaction scale $\rho_N > 0$ are the same quantities that appear in Definition 2.1. The regularized Gaussian interaction kernel at level N is

$$K_{\rho_N}(x, y) := \kappa_0 + \exp\left(-\frac{|x - y|^2}{2\rho_N^2}\right), \quad x, y \in \mathbb{R}^3.$$

Remark 3.2. The parameters in Definition 3.1 are the finite-data counterparts of the kernel and color parameters used in the current lattice paper. In the present replacement paper they are stated explicitly so that the only external dependence is on the convergence statements of `01_convergence`.

Definition 3.3 (Recorded trajectory data). Fix $N \geq 2$ and a finite observation horizon $T \in \mathbb{N}$. For each particle index $i \in \{1, \dots, N\}$ and each time slice $t \in \{0, \dots, T\}$, let

$$X_i^{(N)}(t) \in \mathbb{R}^3, \quad V_i^{(N)}(t) \in \mathbb{R}^3$$

be the recorded position and velocity of particle i at time t . For each t , let

$$\mathcal{P}_t^{(N)} \subset \{(i, j) : 1 \leq i, j \leq N, i \neq j\}$$

be the ordered companion relation at time slice t . The finite trajectory record is

$$\mathcal{Q}^{(N)} := \left(\{X_i^{(N)}(t), V_i^{(N)}(t)\}_{1 \leq i \leq N, 0 \leq t \leq T}, \{\mathcal{P}_t^{(N)}\}_{t=0}^T \right).$$

Definition 3.4 (Node set and flat spacetime embedding). For each particle-time pair (i, t) , define a node

$$n_{i,t}^{(N)}.$$

The node set is

$$\mathcal{N}^{(N)} := \{n_{i,t}^{(N)} : 1 \leq i \leq N, 0 \leq t \leq T\}.$$

The flat spacetime embedding is

$$\Xi^{(N)} : \mathcal{N}^{(N)} \rightarrow \mathbb{R}^4, \quad \Xi^{(N)}(n_{i,t}^{(N)}) := (t\tau_N, X_i^{(N)}(t)).$$

Definition 3.5 (Directed edge set). The temporal links are

$$e_{i,t}^{\text{temp}} := (n_{i,t}^{(N)} \rightarrow n_{i,t+1}^{(N)}), \quad 1 \leq i \leq N, 0 \leq t < T.$$

The interaction links are

$$e_{ij,t}^{\text{int}} := (n_{i,t}^{(N)} \rightarrow n_{j,t}^{(N)}), \quad (i, j) \in \mathcal{P}_t^{(N)}, 0 \leq t \leq T.$$

The full directed edge set is

$$\mathcal{E}^{(N)} := \left\{ e_{i,t}^{\text{temp}} : 1 \leq i \leq N, 0 \leq t < T \right\} \cup \left\{ e_{ij,t}^{\text{int}} : (i, j) \in \mathcal{P}_t^{(N)}, 0 \leq t \leq T \right\},$$

and the finite QFT lattice is

$$\mathcal{L}^{(N)} := (\mathcal{N}^{(N)}, \mathcal{E}^{(N)}).$$

Definition 3.6 (Canonical plaquettes). For each time slice $0 \leq t < T$ and each ordered pair

$$(i, j) \in \mathcal{P}_t^{(N)} \cap \mathcal{P}_{t+1}^{(N)},$$

define the canonical elementary plaquette

$$P_{ij,t}^{(N)} := (n_{i,t}^{(N)} \rightarrow n_{i,t+1}^{(N)} \rightarrow n_{j,t+1}^{(N)} \rightarrow n_{j,t}^{(N)} \rightarrow n_{i,t}^{(N)}).$$

The family of all canonical plaquettes is

$$\mathcal{C}_{\square}^{(N)} := \left\{ P_{ij,t}^{(N)} : (i, j) \in \mathcal{P}_t^{(N)} \cap \mathcal{P}_{t+1}^{(N)}, 0 \leq t < T \right\}.$$

Definition 3.7 (Plaquette geometry). For $P = P_{ij,t}^{(N)} \in \mathcal{C}_{\square}^{(N)}$, define the base point

$$x_P^{(N)} := \Xi^{(N)}(n_{i,t}^{(N)}) \in \mathbb{R}^4,$$

the temporal displacement

$$\tau_{i,t}^{(N)} := \Xi^{(N)}(n_{i,t+1}^{(N)}) - \Xi^{(N)}(n_{i,t}^{(N)}),$$

the interaction displacement

$$\sigma_{ij,t}^{(N)} := \Xi^{(N)}(n_{j,t}^{(N)}) - \Xi^{(N)}(n_{i,t}^{(N)}),$$

the oriented area bivector

$$\Sigma_P^{(N),\mu\nu} := \tau_{i,t}^{(N),\mu} \sigma_{ij,t}^{(N),\nu} - \tau_{i,t}^{(N),\nu} \sigma_{ij,t}^{(N),\mu},$$

and the plaquette area

$$A_P^{(N)} := \left(\frac{1}{2} \sum_{\mu,\nu=0}^3 \Sigma_P^{(N),\mu\nu} \Sigma_P^{(N),\mu\nu} \right)^{1/2}.$$

Remark 3.8. Definitions 3.4–3.7 are the full finite QFT lattice needed for the continuum route. They are now stated inside the present paper rather than imported by citation.

4 Finite gauge data on the QFT lattice

4.1 Companion weights and site colors

Definition 4.1 (Companion weights). For each node $n_{i,t}^{(N)} \in \mathcal{N}^{(N)}$, define the raw interaction degree

$$\deg_{\text{link}}^{(N)}(i, t) := \sum_{(i,j) \in \mathcal{P}_t^{(N)}} K_{\rho_N}(X_i^{(N)}(t), X_j^{(N)}(t)),$$

and the normalized companion weights

$$\omega_{ij}^{(N)}(t) := \begin{cases} \frac{K_{\rho_N}(X_i^{(N)}(t), X_j^{(N)}(t))}{\deg_{\text{link}}^{(N)}(i, t)}, & (i, j) \in \mathcal{P}_t^{(N)} \text{ and } \deg_{\text{link}}^{(N)}(i, t) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 4.2 (Normalization of the companion weights). *For every N , particle index i , and time slice t ,*

$$\omega_{ij}^{(N)}(t) \geq 0 \quad \text{for all } j,$$

and

$$\sum_{j:(i,j) \in \mathcal{P}_t^{(N)}} \omega_{ij}^{(N)}(t) = \mathbf{1}_{\{\deg_{\text{link}}^{(N)}(i,t) > 0\}}.$$

Proof. Nonnegativity follows from $K_{\rho_N} \geq \kappa_0 > 0$. If $\deg_{\text{link}}^{(N)}(i,t) > 0$, then by direct summation of the definition,

$$\sum_{j:(i,j) \in \mathcal{P}_t^{(N)}} \omega_{ij}^{(N)}(t) = \frac{1}{\deg_{\text{link}}^{(N)}(i,t)} \sum_{j:(i,j) \in \mathcal{P}_t^{(N)}} K_{\rho_N}(X_i^{(N)}(t), X_j^{(N)}(t)) = 1.$$

If $\deg_{\text{link}}^{(N)}(i,t) = 0$, each summand is zero by definition. $\square$

Definition 4.3 (Viscous force on interaction links and nodes). For each interaction link

$$e_{ij,t}^{\text{int}} = (n_{i,t}^{(N)} \rightarrow n_{j,t}^{(N)}) \in \mathcal{E}^{(N)},$$

define the pair force

$$\mathbf{F}_{\text{link}}^{(N)}(e_{ij,t}^{\text{int}}) := \nu_{\text{visc}} \omega_{ij}^{(N)}(t) (V_j^{(N)}(t) - V_i^{(N)}(t)) \in \mathbb{R}^3.$$

The site force at $x = n_{i,t}^{(N)}$ is the outgoing interaction sum

$$\mathbf{F}_{\text{visc}}^{(N)}(x) := \sum_{(i,j) \in \mathcal{P}_t^{(N)}} \mathbf{F}_{\text{link}}^{(N)}(e_{ij,t}^{\text{int}}).$$

Definition 4.4 (Color triplets). For each site $x = n_{i,t}^{(N)}$ and each component $\alpha \in \{1, 2, 3\}$, define the unnormalized color component

$$\tilde{c}_x^{(N),\alpha} := (\mathbf{F}_{\text{visc}}^{(N)}(x))_\alpha \exp\left(i \frac{\mu_p V_i^{(N),\alpha}(t) \ell_0}{\hbar_{\text{eff}}}\right),$$

the unnormalized color vector

$$\tilde{\mathbf{c}}_x^{(N)} := (\tilde{c}_x^{(N),1}, \tilde{c}_x^{(N),2}, \tilde{c}_x^{(N),3}) \in \mathbb{C}^3,$$

and the normalized color triplet

$$\mathbf{c}_x^{(N)} := \frac{\tilde{\mathbf{c}}_x^{(N)}}{\sqrt{(\tilde{\mathbf{c}}_x^{(N)})^\dagger \tilde{\mathbf{c}}_x^{(N)} + \epsilon_c^2}} \in \mathbb{C}^3.$$

Lemma 4.5 (Color normalization). *For every site $x \in \mathcal{N}^{(N)}$,*

$$(\mathbf{c}_x^{(N)})^\dagger \mathbf{c}_x^{(N)} = \frac{(\tilde{\mathbf{c}}_x^{(N)})^\dagger \tilde{\mathbf{c}}_x^{(N)}}{(\tilde{\mathbf{c}}_x^{(N)})^\dagger \tilde{\mathbf{c}}_x^{(N)} + \epsilon_c^2} \in [0, 1).$$

In particular, the normalized color vector is well-defined and obeys

$$\|\mathbf{c}_x^{(N)}\| < 1.$$

Proof. The denominator is strictly positive because $\epsilon_c > 0$. The displayed identity is immediate from the definition. Since $r/(r + \epsilon_c^2) \in [0, 1)$ for every $r \geq 0$, the claim follows. $\square$

4.2 Link variables and parallel transport

Definition 4.6 (Gell-Mann basis and Hermitian generator space). The Gell-Mann matrices are

$$\begin{aligned}\lambda_1 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda_2 &= \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda_3 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ \lambda_4 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, & \lambda_5 &= \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, & \lambda_6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \\ \lambda_7 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, & \lambda_8 &= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.\end{aligned}$$

Each λ_a is Hermitian and traceless, and

$$\text{tr}(\lambda_a \lambda_b) = 2\delta_{ab}.$$

We write

$$\mathfrak{h}_3 := \{H \in \mathbb{C}^{3 \times 3} : H^\dagger = H, \text{ tr } H = 0\}$$

for the Hermitian traceless generator space. The Lie algebra $\mathfrak{su}(3)$ is identified with $i\mathfrak{h}_3$ in the usual physics convention.

Definition 4.7 (Edge generators and link transports). For each oriented edge $e = (x \rightarrow y) \in \mathcal{E}^{(N)}$, define

$$\begin{aligned}\mathcal{A}_e^{(N)} &:= \frac{1}{2} \sum_{a=1}^8 \Re((\mathbf{c}_x^{(N)})^\dagger \lambda_a (\mathbf{c}_y^{(N)} - \mathbf{c}_x^{(N)})) \lambda_a \in \mathfrak{h}_3, \\ \Theta_e^{(N)} &:= g_3 \mathcal{A}_e^{(N)} \in \mathfrak{h}_3,\end{aligned}$$

and

$$U_3^{(N)}(e) := \exp(i\Theta_e^{(N)}) \in SU(3).$$

For the reversed orientation, set

$$U_3^{(N)}(e^{-1}) := (U_3^{(N)}(e))^{-1}.$$

Lemma 4.8 (Link transport lies in $SU(3)$). *For every oriented edge $e \in \mathcal{E}^{(N)}$, the matrix $U_3^{(N)}(e)$ belongs to $SU(3)$.*

Proof. Each coefficient

$$\Re((\mathbf{c}_x^{(N)})^\dagger \lambda_a (\mathbf{c}_y^{(N)} - \mathbf{c}_x^{(N)}))$$

is real, and each λ_a is Hermitian and traceless. Hence $\Theta_e^{(N)}$ is Hermitian and traceless. Therefore $i\Theta_e^{(N)}$ is skew-Hermitian and traceless, so

$$U_3^{(N)}(e) = \exp(i\Theta_e^{(N)})$$

is unitary with determinant

$$\det U_3^{(N)}(e) = \exp(i \text{tr } \Theta_e^{(N)}) = 1.$$

□

4.3 Plaquette field strength and Wilson action

Definition 4.9 (Gauge maps and transformed links). A gauge map at level N is a function

$$\Omega^{(N)} : \mathcal{N}^{(N)} \rightarrow SU(3).$$

Its action on site colors and links is

$$\begin{aligned} \mathbf{c}_x^{(N),\Omega} &:= \Omega^{(N)}(x) \mathbf{c}_x^{(N)}, \\ U_3^{(N),\Omega}(x \rightarrow y) &:= \Omega^{(N)}(y) U_3^{(N)}(x \rightarrow y) (\Omega^{(N)}(x))^{-1}. \end{aligned}$$

Definition 4.10 (Plaquette holonomy, discrete field strength, and Wilson action). For $P = P_{ij,t}^{(N)} \in \mathcal{C}_{\square}^{(N)}$, define the plaquette holonomy

$$U_3^{(N)}(P) := U_3^{(N)}(n_{i,t}^{(N)} \rightarrow n_{i,t+1}^{(N)}) U_3^{(N)}(n_{i,t+1}^{(N)} \rightarrow n_{j,t+1}^{(N)}) U_3^{(N)}(n_{j,t+1}^{(N)} \rightarrow n_{j,t}^{(N)}) U_3^{(N)}(n_{j,t}^{(N)} \rightarrow n_{i,t}^{(N)}).$$

Whenever $A_P^{(N)} > 0$ and $U_3^{(N)}(P)$ lies in the principal logarithm domain, define the discrete field strength

$$F_P^{(N)} := \frac{1}{ig_3 A_P^{(N)}} \log U_3^{(N)}(P) \in \mathfrak{h}_3.$$

The Wilson action is

$$S_W^{(N)} := \frac{6}{g_3^2} \sum_{P \in \mathcal{C}_{\square}^{(N)}} \left(1 - \frac{1}{3} \Re \operatorname{tr} U_3^{(N)}(P) \right).$$

Proposition 4.11 (Exact plaquette exponential identity and Wilson expansion). *Let $P \in \mathcal{C}_{\square}^{(N)}$ be such that $A_P^{(N)} > 0$ and $U_3^{(N)}(P)$ lies in the principal logarithm domain. Then*

$$U_3^{(N)}(P) = \exp(ig_3 A_P^{(N)} F_P^{(N)}).$$

If in addition

$$\|g_3 A_P^{(N)} F_P^{(N)}\|_{\text{op}} \leq \delta$$

for some fixed $\delta > 0$, then there exists a constant $C_\delta < \infty$, depending only on δ and the matrix norm convention, such that

$$1 - \frac{1}{3} \Re \operatorname{tr} U_3^{(N)}(P) = \frac{g_3^2 (A_P^{(N)})^2}{6} \operatorname{tr}((F_P^{(N)})^2) + R_P^{(N)},$$

with

$$|R_P^{(N)}| \leq C_\delta g_3^3 (A_P^{(N)})^3 \|F_P^{(N)}\|_{\text{op}}^3.$$

Proof. By Definition 4.10,

$$F_P^{(N)} = \frac{1}{ig_3 A_P^{(N)}} \log U_3^{(N)}(P),$$

so multiplication by $ig_3 A_P^{(N)}$ gives

$$\log U_3^{(N)}(P) = ig_3 A_P^{(N)} F_P^{(N)}.$$

Exponentiating both sides yields the exact identity

$$U_3^{(N)}(P) = \exp(ig_3 A_P^{(N)} F_P^{(N)}).$$

Set

$$X_P^{(N)} := ig_3 A_P^{(N)} F_P^{(N)}.$$

Because $F_P^{(N)} \in \mathfrak{h}_3$, the matrix $X_P^{(N)}$ is skew-Hermitian and traceless. The exponential series gives

$$e^{X_P^{(N)}} = I + X_P^{(N)} + \frac{1}{2}(X_P^{(N)})^2 + \mathcal{R}_P^{(N)},$$

where the third-order remainder satisfies the standard matrix bound

$$\|\mathcal{R}_P^{(N)}\|_{\text{op}} \leq C_\delta \|X_P^{(N)}\|_{\text{op}}^3$$

whenever $\|X_P^{(N)}\|_{\text{op}} \leq \delta$. Taking traces and using $\text{tr } X_P^{(N)} = 0$, we obtain

$$\text{tr } U_3^{(N)}(P) = 3 + \frac{1}{2} \text{tr}((X_P^{(N)})^2) + \text{tr } \mathcal{R}_P^{(N)}.$$

Since $(ig_3 A_P^{(N)})^2 = -g_3^2 (A_P^{(N)})^2$,

$$\text{tr}((X_P^{(N)})^2) = -g_3^2 (A_P^{(N)})^2 \text{tr}((F_P^{(N)})^2).$$

Therefore

$$\Re \text{tr } U_3^{(N)}(P) = 3 - \frac{g_3^2 (A_P^{(N)})^2}{2} \text{tr}((F_P^{(N)})^2) + \Re \text{tr } \mathcal{R}_P^{(N)}.$$

Subtracting from 1 after division by 3 gives

$$1 - \frac{1}{3} \Re \text{tr } U_3^{(N)}(P) = \frac{g_3^2 (A_P^{(N)})^2}{6} \text{tr}((F_P^{(N)})^2) - \frac{1}{3} \Re \text{tr } \mathcal{R}_P^{(N)}.$$

It remains to estimate the last term. In fixed dimension, $|\text{tr } Y| \leq 3\|Y\|_{\text{op}}$ for every 3×3 matrix Y , so

$$\left| -\frac{1}{3} \Re \text{tr } \mathcal{R}_P^{(N)} \right| \leq \|\mathcal{R}_P^{(N)}\|_{\text{op}} \leq C_\delta \|X_P^{(N)}\|_{\text{op}}^3.$$

Since $\|X_P^{(N)}\|_{\text{op}} = g_3 A_P^{(N)} \|F_P^{(N)}\|_{\text{op}}$, the claimed bound follows. $\square$

4.4 Gauge invariance

Proposition 4.12 (Finite gauge invariance of plaquette traces). *For every plaquette $P \in \mathcal{C}_\square^{(N)}$ and every gauge map $\Omega^{(N)}$,*

$$\text{tr } U_3^{(N),\Omega}(P) = \text{tr } U_3^{(N)}(P).$$

Consequently,

$$S_W^{(N),\Omega} = S_W^{(N)}.$$

Proof. Write the plaquette as the closed four-step loop

$$v_0 \rightarrow v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_4 = v_0.$$

Under the gauge action from Definition 4.9,

$$U_3^{(N),\Omega}(P) = \prod_{k=1}^4 \Omega^{(N)}(v_k) U_3^{(N)}(v_{k-1} \rightarrow v_k) (\Omega^{(N)}(v_{k-1}))^{-1}.$$

All consecutive inner factors telescope, leaving

$$U_3^{(N),\Omega}(P) = \Omega^{(N)}(v_0) U_3^{(N)}(P) (\Omega^{(N)}(v_0))^{-1}.$$

Trace cyclicity gives the first claim, and summing over all plaquettes gives the Wilson-action invariance. $\square$

Remark 4.13. At this point the present paper contains the full finite QFT lattice and gauge construction needed for the continuum route. The only external theorem input still intended to come from `01_convergence` is the convergence and mean-field backbone, not the finite lattice gauge data themselves.

Definition 4.14 (Compact-local big- O notation). Let $\mathcal{K} \subset \mathbb{R}^4$ be compact and let $m \geq 0$. For a sequence of scalar, vector, or matrix-valued functions R_N defined on $\mathcal{K}$, we write

$$R_N = O_{\mathcal{K}}(a_N^m)$$

if there exists a constant $C_{\mathcal{K}} < \infty$, independent of N , such that

$$\sup_{x \in \mathcal{K}} |R_N(x)| \leq C_{\mathcal{K}} a_N^m$$

for all sufficiently large N . For quantities attached to plaquettes or edges with base point or midpoint in $\mathcal{K}$, the same notation means a uniform bound over those plaquettes or edges.

5 Translation dictionary from the old route

6 Hypothesis package for the local continuum route

The present section is a direct translation of the proof stack used in the old Yang–Mills chapter. No alternative localization theorem is introduced here. The logical order is exactly the old one:

1. small-field expansion on each plaquette;
2. plaquette area control at scale $\rho_N \tau_N$;
3. Riemann-sum convergence of the weighted plaquette density;
4. global remainder control;
5. coupling identification.

Old document	4_ymmg translation
temporal spacing τ	temporal scale τ_N
localization spacing ρ	interaction scale ρ_N
common lattice spacing a	$a_N := \max\{\tau_N, \rho_N\}$
plaquette P	canonical plaquette $P_{ij,t}^{(N)} \in \mathcal{C}_{\square}^{(N)}$
plaquette area A_P	$A_P^{(N)} = \Sigma_P^{(N)} $
holonomy U_P	$U_3^{(N)}(P)$
field strength F_P	$F_P^{(N)} = (ig_3 A_P^{(N)})^{-1} \log U_3^{(N)}(P)$
Wilson action $S_{\text{YM}}^{\text{disc}}$	$S_W^{(N)}$
continuum action $\frac{1}{4} \int F^2$	local $SU(3)$ Yang–Mills action on $\mathbb{R}^4$

Table 1: Dictionary from the old continuum-limit route to the present notation.

Assumption 6.1 (Flat refinement regime). There exist positive sequences τ_N, ρ_N such that

$$\tau_N \downarrow 0, \quad \rho_N \downarrow 0, \quad a_N := \max\{\tau_N, \rho_N\} \downarrow 0.$$

There exist constants $0 < c_{\text{ar}} \leq C_{\text{ar}} < \infty$ such that

$$c_{\text{ar}} \leq \frac{\tau_N}{\rho_N} \leq C_{\text{ar}} \quad \text{for all } N \geq N_0.$$

There also exist constants $0 < c_A \leq C_A < \infty$ such that

$$c_A \rho_N \tau_N \leq A_P^{(N)} \leq C_A \rho_N \tau_N \quad \text{for all } P \in \mathcal{C}_{\square}^{(N)}.$$

Finally the total number of canonical plaquettes satisfies the four-dimensional counting law

$$|\mathcal{C}_{\square}^{(N)}| = O(a_N^{-4}).$$

Definition 6.2 (Flat-branch continuum curvature). Let

$$F_{\mu\nu} : \mathbb{R}^4 \rightarrow \mathfrak{h}_3$$

be an antisymmetric matrix-valued two-form on the flat Euclidean background, expanded in the Gell-Mann basis as

$$F_{\mu\nu}(x) = \frac{1}{2} \sum_{a=1}^8 F_{\mu\nu}^a(x) \lambda_a.$$

Assumption 6.3 (Small-field holonomy regime and continuum field matching). There exists a curvature field

$$F_{\mu\nu} \in C_{\text{loc}}^{\alpha}(\mathbb{R}^4; \mathfrak{h}_3) \quad \text{for some } \alpha > 0,$$

such that

$$x \mapsto \text{tr}(F_{\mu\nu}(x) F_{\mu\nu}(x)) \in C_c(\mathbb{R}^4).$$

For every compact set $\mathcal{K} \subset \mathbb{R}^4$, there exist constants $\delta_{\mathcal{K}} > 0$, $C_{\mathcal{K}} < \infty$, and $N_{\mathcal{K}} \in \mathbb{N}$ such that for all $N \geq N_{\mathcal{K}}$ and all plaquettes $P \in \mathcal{C}_{\square}^{(N)}$ with $x_P^{(N)} \in \mathcal{K}$ one has:

(i) the plaquette holonomy lies in the principal logarithm domain and

$$\|g_3 A_P^{(N)} F_P^{(N)}\|_{\text{op}} \leq \delta_{\mathcal{K}};$$

(ii) the local field-strength matching estimate

$$\mathrm{tr}(F_P^{(N)} F_P^{(N)}) = \mathrm{tr}(F_{\mu\nu}(x_P^{(N)}) F_{\mu\nu}(x_P^{(N)})) + O_K(a_N^\alpha)$$

holds uniformly in P .

Assumption 6.4 (Weighted plaquette Riemann sums). For every compactly supported continuous test function $\varphi \in C_c(\mathbb{R}^4)$,

$$\sum_{P \in \mathcal{C}_\square^{(N)}} (A_P^{(N)})^2 \varphi(x_P^{(N)}) \longrightarrow \int_{\mathbb{R}^4} \varphi(x) \, dx.$$

Equivalently, the weighted plaquette measures

$$\Gamma_\square^{(N)} := \sum_{P \in \mathcal{C}_\square^{(N)}} (A_P^{(N)})^2 \delta_{x_P^{(N)}}$$

converge weakly to Lebesgue measure on $\mathbb{R}^4$.

Remark 6.5. Assumption 6.4 is the literal translated form of the old sentence that the Wilson sum is a Riemann sum for the Euclidean volume form. It is kept in scalar weighted-measure form because that is the form used by the original proof.

7 Translated continuum-limit theorem

Definition 7.1 (Continuum Yang–Mills action in the present convention). For the curvature field of Definition 6.2, define

$$S_{\mathrm{YM}}^{\mathrm{cont}}[F] := \frac{1}{4} \int_{\mathbb{R}^4} \sum_{a=1}^8 F_{\mu\nu}^a(x) F_{\mu\nu}^a(x) \, dx.$$

Equivalently, because

$$\mathrm{tr}(F_{\mu\nu}(x) F_{\mu\nu}(x)) = \frac{1}{2} \sum_{a=1}^8 F_{\mu\nu}^a(x) F_{\mu\nu}^a(x),$$

one has

$$S_{\mathrm{YM}}^{\mathrm{cont}}[F] = \frac{1}{2} \int_{\mathbb{R}^4} \mathrm{tr}(F_{\mu\nu}(x) F_{\mu\nu}(x)) \, dx.$$

Theorem 7.2 (Continuum limit of the trajectory-induced Yang–Mills action). *Assume the translated old-route hypotheses Assumptions 6.1, 6.3, and 6.4. Assume also that the continuum and lattice couplings are identified by*

$$g_{\mathrm{phys}} = g_3.$$

Set the finite Wilson prefactor to the standard four-dimensional $SU(3)$ value

$$\beta_N := \frac{6}{g_3^2},$$

and define

$$S_{\mathrm{YM}}^{\mathrm{disc},(N)} := \beta_N \sum_{P \in \mathcal{C}_\square^{(N)}} \left(1 - \frac{1}{3} \Re \mathrm{tr} U_3^{(N)}(P) \right).$$

Then

$$S_{\text{YM}}^{\text{disc},(N)} \longrightarrow \frac{1}{4} \int_{\mathbb{R}^4} \sum_{a=1}^8 F_{\mu\nu}^a(x) F_{\mu\nu}^a(x) \, dx = S_{\text{YM}}^{\text{cont}}[F].$$

Proof. The proof is the exact old five-step continuum argument translated into the present notation.

Step 1: Small-field expansion. For each plaquette $P \in \mathcal{C}_{\square}^{(N)}$, set

$$s_P^{(N)} := 1 - \frac{1}{3} \Re \operatorname{tr} U_3^{(N)}(P).$$

By Assumption 6.3, the plaquette holonomy lies in the principal logarithm domain on each fixed compact set for all sufficiently large N , and its logarithm obeys the small-field bound required by Proposition 4.11. Hence

$$s_P^{(N)} = \frac{g_3^2 (A_P^{(N)})^2}{6} \operatorname{tr}(F_P^{(N)} F_P^{(N)}) + R_P^{(N)}.$$

Moreover, for plaquettes with base point in a fixed compact set $\mathcal{K}$, the same assumption yields

$$\operatorname{tr}(F_P^{(N)} F_P^{(N)}) = \operatorname{tr}(F_{\mu\nu}(x_P^{(N)}) F_{\mu\nu}(x_P^{(N)})) + O_{\mathcal{K}}(a_N^\alpha).$$

Step 2: Plaquette area control. Assumption 6.1 gives

$$c_A \rho_N \tau_N \leq A_P^{(N)} \leq C_A \rho_N \tau_N$$

uniformly in P . Since the aspect ratio τ_N/ρ_N is bounded above and below, this implies

$$A_P^{(N)} = O(a_N^2)$$

uniformly over $\mathcal{C}_{\square}^{(N)}$. The same assumption gives the plaquette counting law

$$|\mathcal{C}_{\square}^{(N)}| = O(a_N^{-4}).$$

Fix a compact set $\mathcal{K}$ containing the support of the compactly supported continuum density

$$\varphi(x) := \operatorname{tr}(F_{\mu\nu}(x) F_{\mu\nu}(x)).$$

Because $\varphi \in C_c(\mathbb{R}^4)$ by Assumption 6.3, the field $F_{\mu\nu}$ is bounded on $\mathcal{K}$, and the same assumption implies that the discrete field strengths are uniformly bounded on plaquettes with $x_P^{(N)} \in \mathcal{K}$ once N is large. Combining this boundedness with Proposition 4.11 and the estimate $A_P^{(N)} = O(a_N^2)$ yields

$$|R_P^{(N)}| = O(a_N^6) \quad \text{for every plaquette with } x_P^{(N)} \in \mathcal{K}.$$

Step 3: Riemann-sum convergence. Summing the small-field expansion and multiplying by the Wilson prefactor $\beta_N = 6/g_3^2$, we obtain

$$S_{\text{YM}}^{\text{disc},(N)} = \sum_{P \in \mathcal{C}_{\square}^{(N)}} (A_P^{(N)})^2 \operatorname{tr}(F_P^{(N)} F_P^{(N)}) + \beta_N \sum_{P \in \mathcal{C}_{\square}^{(N)}} R_P^{(N)}.$$

Assumption 6.3 identifies the first factor with

$$\operatorname{tr}(F_P^{(N)} F_P^{(N)}) = \operatorname{tr}(F_{\mu\nu}(x_P^{(N)}) F_{\mu\nu}(x_P^{(N)})) + O_{\mathcal{K}}(a_N^\alpha)$$

uniformly on the compact set $\mathcal{K}$. Therefore

$$\sum_{P \in \mathcal{C}_{\square}^{(N)}} (A_P^{(N)})^2 \left| \text{tr}(F_P^{(N)} F_P^{(N)}) - \varphi(x_P^{(N)}) \right| \leq C a_N^\alpha \sum_{P: x_P^{(N)} \in \mathcal{K}} (A_P^{(N)})^2.$$

Because the weighted measures from Assumption 6.4 converge weakly to Lebesgue measure, the weighted mass of $\mathcal{K}$ stays bounded, and the right-hand side tends to zero. Applying Assumption 6.4 to the compactly supported function φ then gives

$$\sum_{P \in \mathcal{C}_{\square}^{(N)}} (A_P^{(N)})^2 \text{tr}(F_P^{(N)} F_P^{(N)}) \longrightarrow \int_{\mathbb{R}^4} \text{tr}(F_{\mu\nu}(x) F_{\mu\nu}(x)) \, dx.$$

Step 4: Remainder control. Because each remainder is $O(a_N^6)$ and the number of plaquettes is $O(a_N^{-4})$, one has

$$\left| \sum_{P \in \mathcal{C}_{\square}^{(N)}} R_P^{(N)} \right| \leq C a_N^{-4} \cdot a_N^6 = O(a_N^2) \longrightarrow 0.$$

Since $\beta_N = 6/g_3^2$ is constant in four dimensions, the same limit holds for the remainder contribution in the action.

Step 5: Coupling identification. Combining the previous steps gives

$$S_{\text{YM}}^{\text{disc},(N)} \longrightarrow \int_{\mathbb{R}^4} \text{tr}(F_{\mu\nu}(x) F_{\mu\nu}(x)) \, dx.$$

By Definition 6.2,

$$F_{\mu\nu}(x) = \frac{1}{2} \sum_{a=1}^8 F_{\mu\nu}^a(x) \lambda_a,$$

and the Gell-Mann orthogonality relation yields

$$\text{tr}(F_{\mu\nu}(x) F_{\mu\nu}(x)) = \frac{1}{2} \sum_{a=1}^8 F_{\mu\nu}^a(x) F_{\mu\nu}^a(x).$$

Therefore

$$\int_{\mathbb{R}^4} \text{tr}(F_{\mu\nu}(x) F_{\mu\nu}(x)) \, dx = \frac{1}{4} \int_{\mathbb{R}^4} \sum_{a=1}^8 F_{\mu\nu}^a(x) F_{\mu\nu}^a(x) \, dx = S_{\text{YM}}^{\text{cont}}[F].$$

This proves the continuum limit. □

8 Discharge plan

The present paper follows the old continuum strategy exactly. The remaining work is therefore to discharge the translated versions of those same five ingredients.

H1. Flat refinement regime

- Statement to discharge: Definition 2.1 and Assumption 6.1.
- Required source: the recorded trajectory family must now carry an explicit refinement schedule $N \mapsto (\tau_N, \rho_N)$.
- Output: a theorem proving

$$\tau_N \downarrow 0, \quad \rho_N \downarrow 0, \quad A_P^{(N)} \sim \rho_N \tau_N, \quad |\mathcal{C}_\square^{(N)}| = O(a_N^{-4}).$$

H2. Small-field expansion

- Statement to discharge: Assumption 6.3.
- Required source: the translated holonomy expansion for the $SU(3)$ plaquette variables together with a compact-local continuum field-matching theorem.
- Output:

(i) principal-log control

$$\|g_3 A_P^{(N)} F_P^{(N)}\|_{\text{op}} \leq \delta \mathcal{K};$$

(ii) compact-local approximation

$$\text{tr}(F_P^{(N)} F_P^{(N)}) = \text{tr}(F_{\mu\nu}(x_P^{(N)}) F_{\mu\nu}(x_P^{(N)})) + O_{\mathcal{K}}(a_N^\alpha);$$

(iii) compact support of the continuum energy density

$$x \mapsto \text{tr}(F_{\mu\nu}(x) F_{\mu\nu}(x)) \in C_c(\mathbb{R}^4).$$

The per-plaquette Taylor expansion is already proved internally in Proposition 4.11.

H3. Weighted plaquette Riemann sums

- Statement to discharge: Assumption 6.4.
- Required source: the flat-branch plaquette sampling law.
- Output:

$$\sum_{P \in \mathcal{C}_\square^{(N)}} (A_P^{(N)})^2 \varphi(x_P^{(N)}) \longrightarrow \int_{\mathbb{R}^4} \varphi(x) \, dx \quad \forall \varphi \in C_c(\mathbb{R}^4).$$

H4. Global remainder control

- Statement to discharge: Step 4 of Theorem 7.2.
- Required source: the combination of the remainder estimate and the plaquette count estimate.
- Output:

$$\sum_{P \in \mathcal{C}_\square^{(N)}} R_P^{(N)} \rightarrow 0.$$

H5. Coupling identification

- Statement to discharge: Step 5 of Theorem 7.2.
- Required source: the component decomposition

$$F_{\mu\nu} = \frac{1}{2} \sum_{a=1}^8 F_{\mu\nu}^a \lambda_a$$

and the standard four-dimensional $SU(3)$ Wilson prefactor

$$\beta_N = \frac{6}{g_3^2}.$$

- Output:

$$\int_{\mathbb{R}^4} \text{tr}(F_{\mu\nu} F_{\mu\nu}) \, dx = \frac{1}{4} \int_{\mathbb{R}^4} \sum_{a=1}^8 F_{\mu\nu}^a F_{\mu\nu}^a \, dx.$$

9 How this paper substitutes the current lattice paper

The finite sections of the current `03_lattice_qft` are now restated inside the present paper, so the theorem chain itself no longer depends on that document. What is replaced is the previous thermodynamic target section. The large- N target of the replacement paper is Theorem 7.2, namely the local Euclidean Yang–Mills action on $\mathbb{R}^4$.

Remark 9.1. Once the theorem stack of the present paper is discharged, the downstream quantization and axiom-verification papers should import Theorem 7.2 as their continuum gauge-sector input.

10 Final checklist

Before the present paper can replace the current thermodynamic route, the following numbered outputs must exist in the series.

1. a flat refinement theorem with

$$A_P^{(N)} \sim \rho_N \tau_N, \quad |\mathcal{C}_{\square}^{(N)}| = O(a_N^{-4});$$

2. a translated $SU(3)$ small-field Wilson expansion on canonical plaquettes;
3. a weighted plaquette Riemann-sum theorem;
4. a global remainder bound for the summed cubic error terms;
5. the coupling-identification identity in the present $SU(3)$ convention;
6. the continuum Yang–Mills limit theorem Theorem 7.2.

This list is the precise new-language form of the old continuum proof stack. No new physics has been inserted. Only the notation has been changed from the Fractal Set variables of the old chapter to the

$$(\mathcal{L}^{(N)}, U_3^{(N)}, \mathcal{C}_{\square}^{(N)}, A_P^{(N)}, x_P^{(N)})$$

language of the present series.